when is it your turn?

digitalisierung und programmierung · input, session 7 agenda

1 why it collapses

2 the clocked way

3 the clock-free way

why it collapses

how do eighty people stay together?

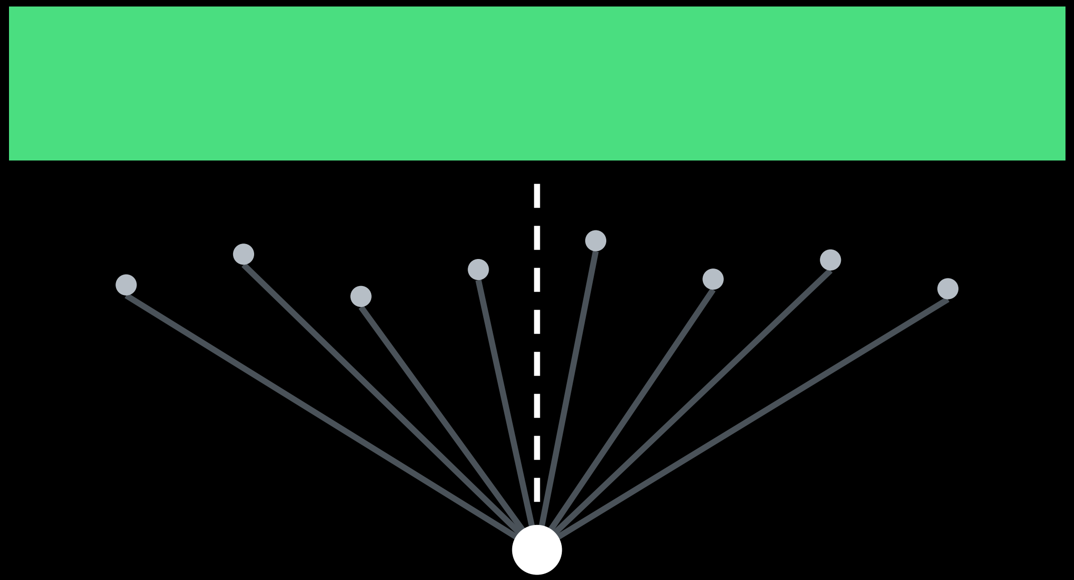
image: ai-generated (gpt-image-2)

your homework: where did it tip?

team	symbols per second	what broke

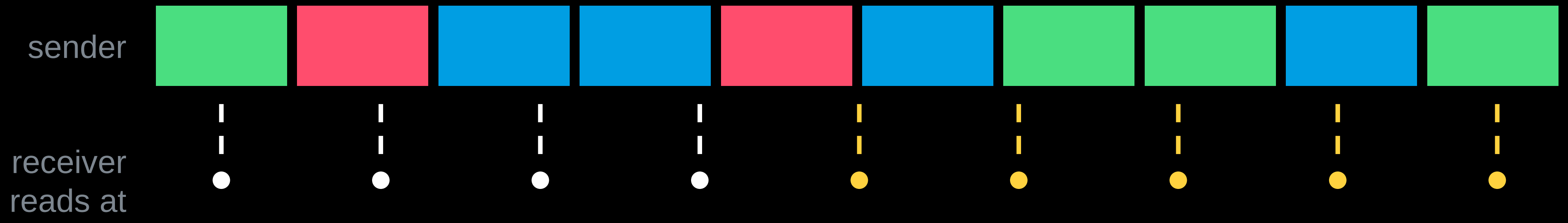
same limit for everyone. very different numbers.

what one dot means on the next slides



one dot in the pictures = many readings, averaged.

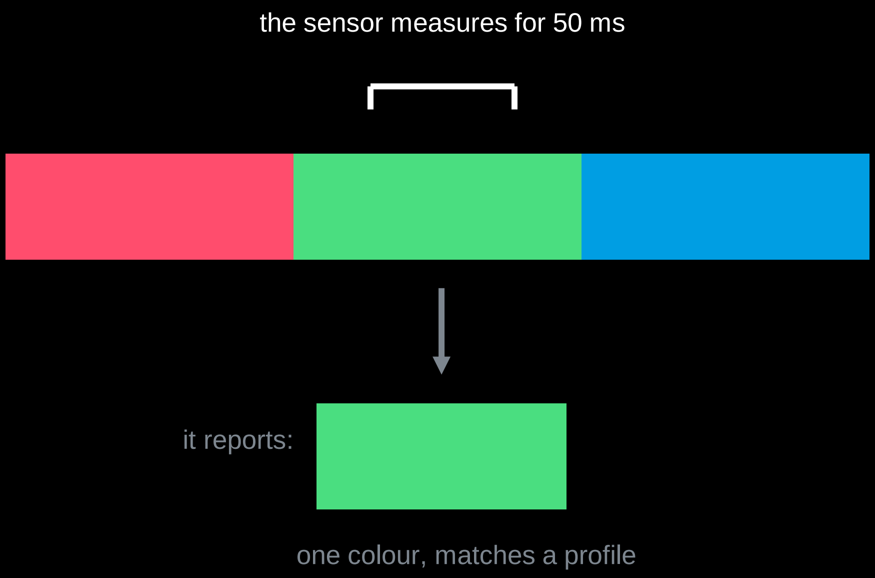
cause one: no two clocks run alike



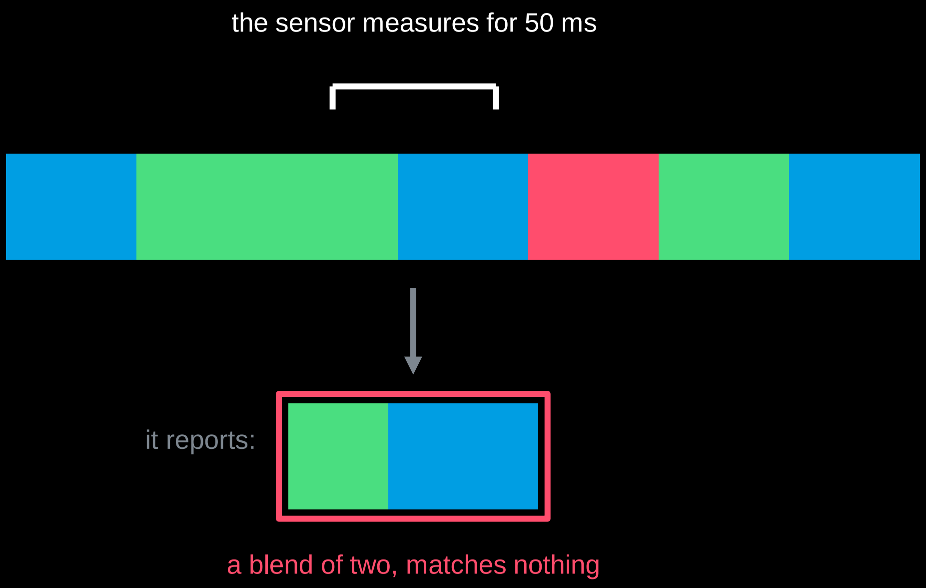
drift adds up. the tiny error per symbol is not the problem, the sum is.

cause two: the sensor needs time to look

symbols 100 ms long



symbols 40 ms long



the sensor averages over its window. no error, no warning, just a meaningless number.

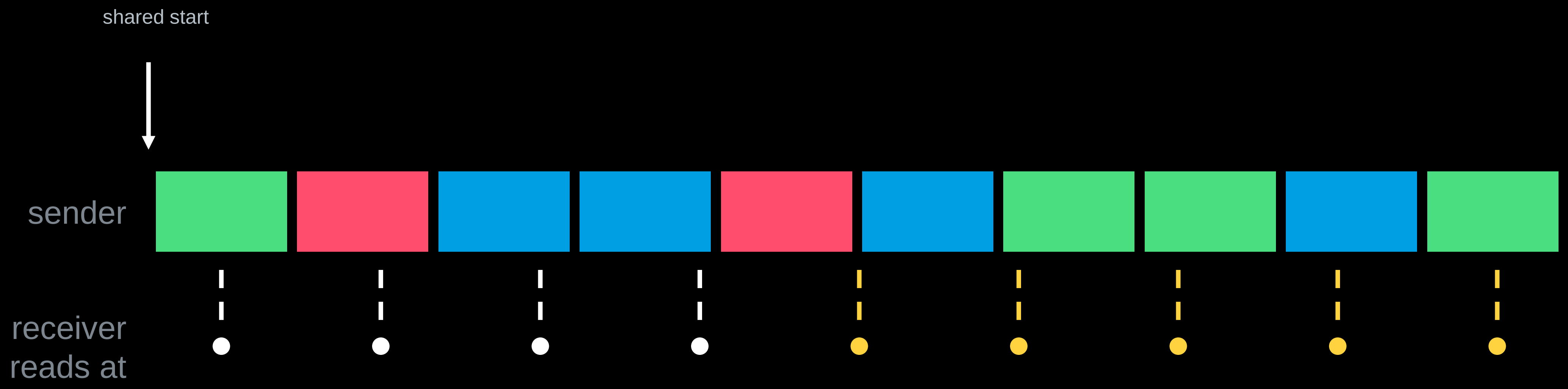
both causes are one question

when does a new symbol begin?

answer one: time tells you.
answer two: change tells you.

the clocked way: time tells you

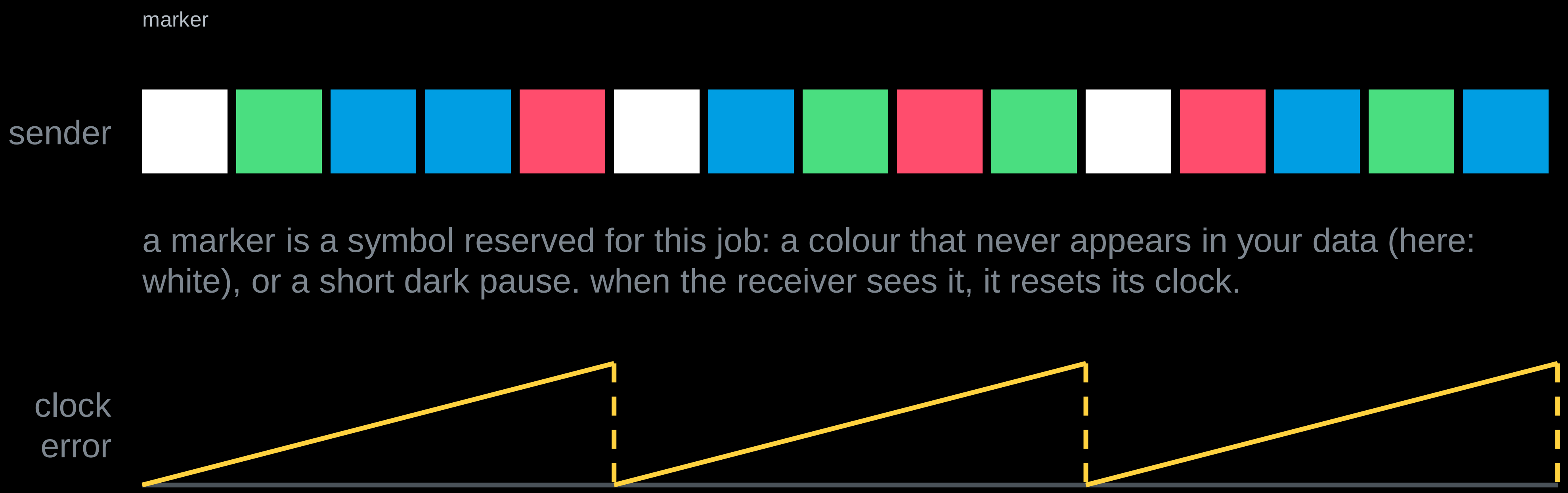
a shared start and a shared rate



0.5 % clock difference at 10 symbols/s: off by one symbol after 20 s

a shared start is not a question of if it tips, only when.

recurring markers reset the drift



a marker is a symbol reserved for this job: a colour that never appears in your data (here: white), or a short dark pause. when the receiver sees it, it resets its clock.

the error can never grow beyond one block.

start once

cheap: not a single symbol spent

drift adds up without limit

recurring markers

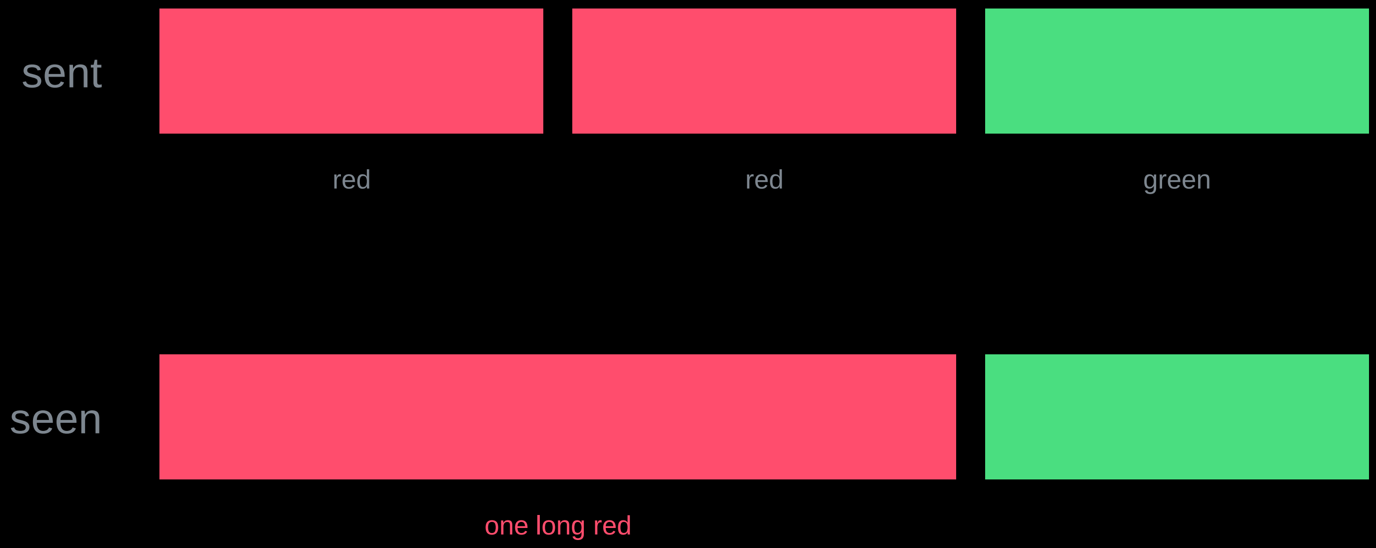
drift stays bounded

markers cost symbols that carry no message

how bad are your clocks? that is a measurement, not a belief.

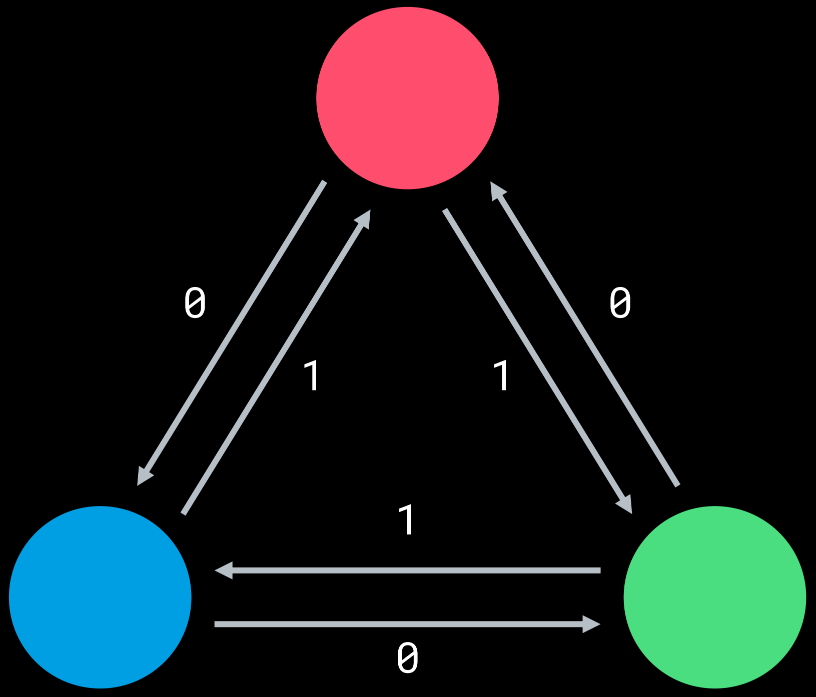
the clock-free way: change tells you

first idea: a change means „next symbol“

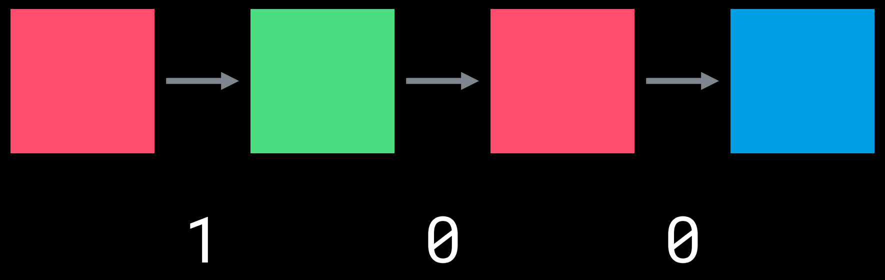


no change, no symbol: „red red“ arrives as one.

the fix: the transition carries the bit



example



measure as often as you like.
decode only on change.

your own colour is never allowed next.

k colours as states		k colours as transitions
	$\log_2(k)$ bits	$\log_2(k-1)$ bits
k = 3	1.58	1.00
k = 4	2.00	1.58
k = 5	2.32	2.00

want a clean 2 bits per transition? you need five colours, not four.

the most common wrong conclusion

clock-free is not time-free

how long must a colour hold to count?
how many transitions per second can the sensor follow?

$$t = \frac{n_{\text{bits}}}{r \cdot b}$$

n bits to send · r symbols per second · b bits per symbol

clocked: $b = \log_2(k)$

clock-free: $b = \log_2(k-1)$

clocked, 16 000 bits, 4 colours, 5 symbols/s: about 27 minutes

two levers, nothing else: bits per symbol, symbols per second.

today

1 clapping experiment

2 workshop: find your maximum rate, with evidence

3 write down which way you lean: clocked or clock-free, and why